

Representation as a Computational Resource in Quantum Compilation

Approximate recoverability tradeoffs and model-specific fault-tolerant consequences

Jinze Yang¹ · Yangyang Li¹ · Xiu-Hao Deng^{2,3}

¹ Xidian University · ² Shenzhen International Quantum Academy · ³ Hefei National Laboratory

01 Compilation destroys structure

Fault-tolerant pipelines deliberately scatter a quantity that was once written in one place — and the scattering is correct, standard, and desirable in each case:

- **Scheduling.** Variational and time-dependent layers split one accumulated phase across many rounds.
- **Synthesis.** An angle becomes a Clifford+ T word in which no single gate carries the angle.
- **Randomized compiling.** Each rotation is twirled into sign-randomized fragments.

We ask what it costs a *downstream* compiler — a cloud transpiler, a control-stack stream optimizer — to put the quantity back together, and show the price is payable in exactly two currencies: **bits of compiler state**, or **magic states**.

SEMANTIC
m records

P_1, θ_1	P_2, θ_2	P_3, θ_3	...	P_m, θ_m
-----------------	-----------------	-----------------	-----	-----------------

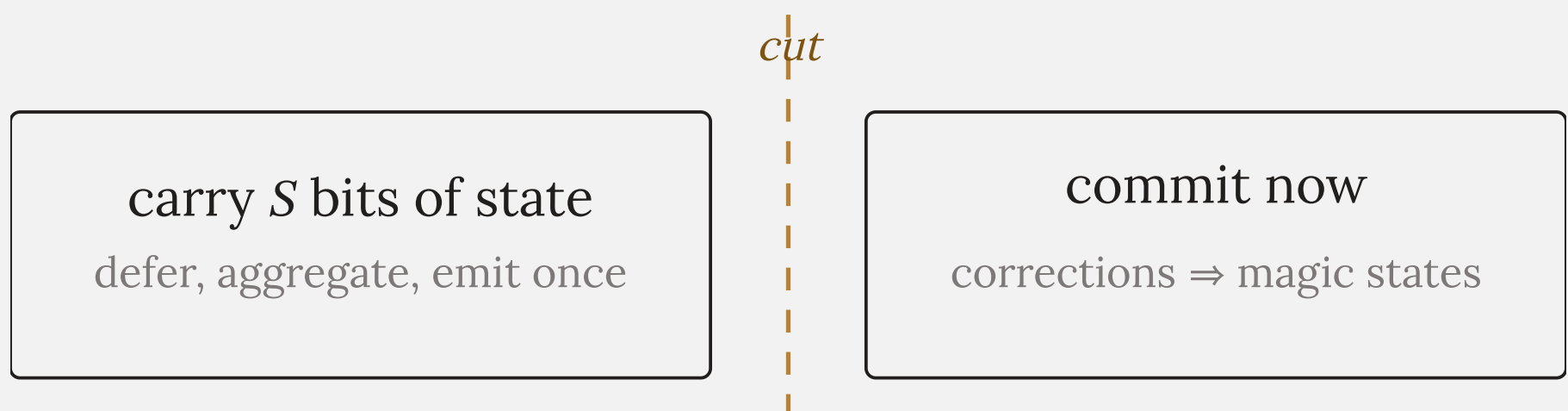
DISPERSED

schedule · twirl · lower

round 1	$\pm a_1$	$\pm a_2$	$\pm a_3$	...	$\pm a_m$
round 2	$\pm a_1$	$\pm a_2$	$\pm a_3$	...	$\pm a_m$
round 3	$\pm a_1$	$\pm a_2$	$\pm a_3$	...	$\pm a_m$

no record — and no prefix — determines the edge total
 $x_j = \sum_i s_j^{(i)} a_j^{(i)} \bmod Q$

AT ANY CUT



One commuting phase layer, lowered into r contiguous rounds of grid residues. At every cut the scattered information is either carried across in state or committed to output and repaid downstream.

02 A budget-explicit model

Every channel that would let a restarted compiler recover the aggregate is charged in bits — no gate count is ever converted into a bit value:

control	store
live window	parameters
intermediate representation	

Recovery is certified, not asserted. The representation-aware pipeline uses bounded recognition, aggregation certificates and conservative fallback; a measured cell becomes numerical only once a symbolic or dense certificate returns `completed_valid`. Unsupported is never conflated with incorrect, and bridge loss is reported apart from compiler quality.

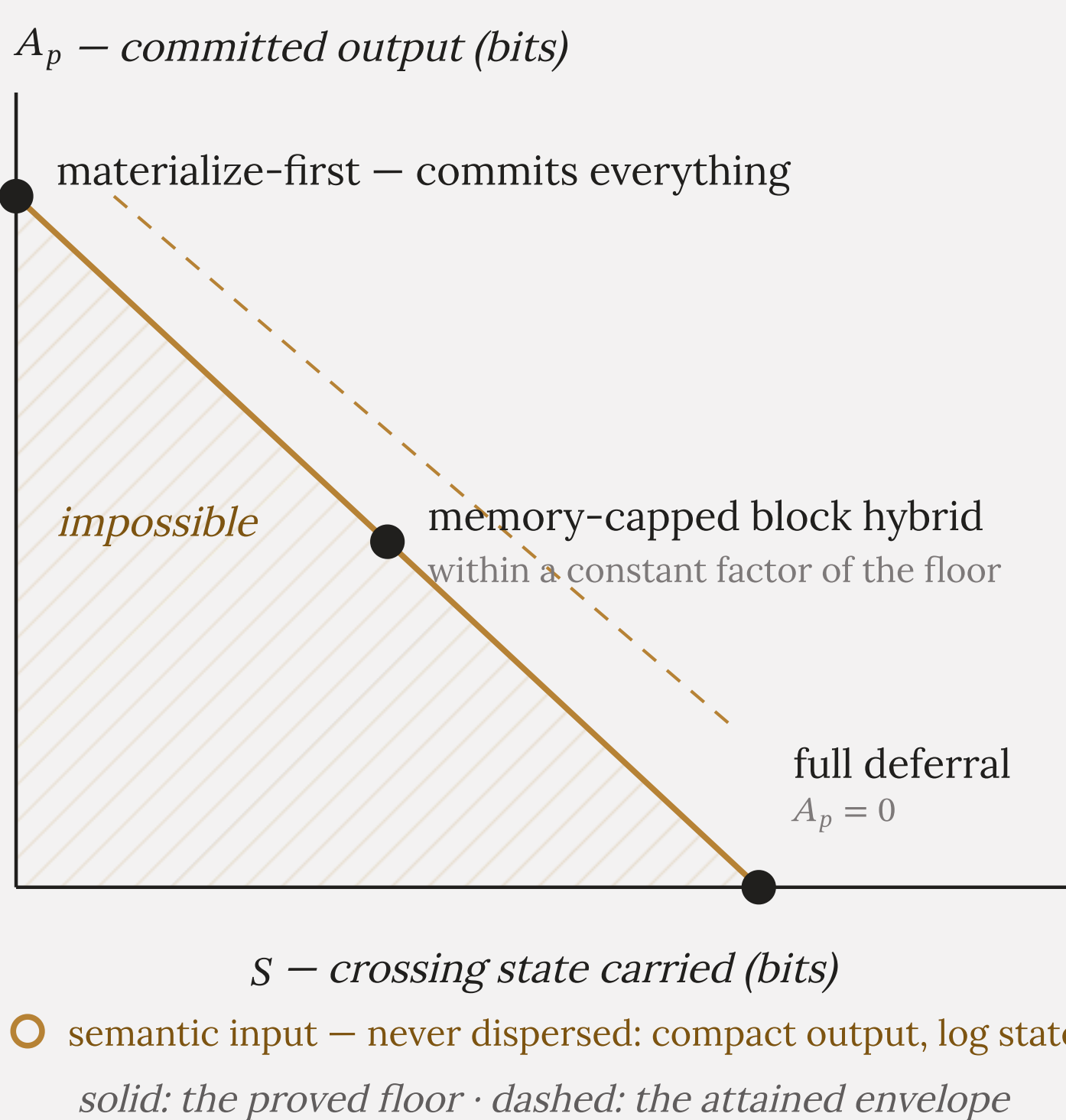
03 The frontier

THEOREM

$$A_p + (2p - 1)S \geq (1 - \delta)m \log_2 K_\epsilon - h_2(\delta)$$

for any compiler making p forward passes and failing with probability at most δ

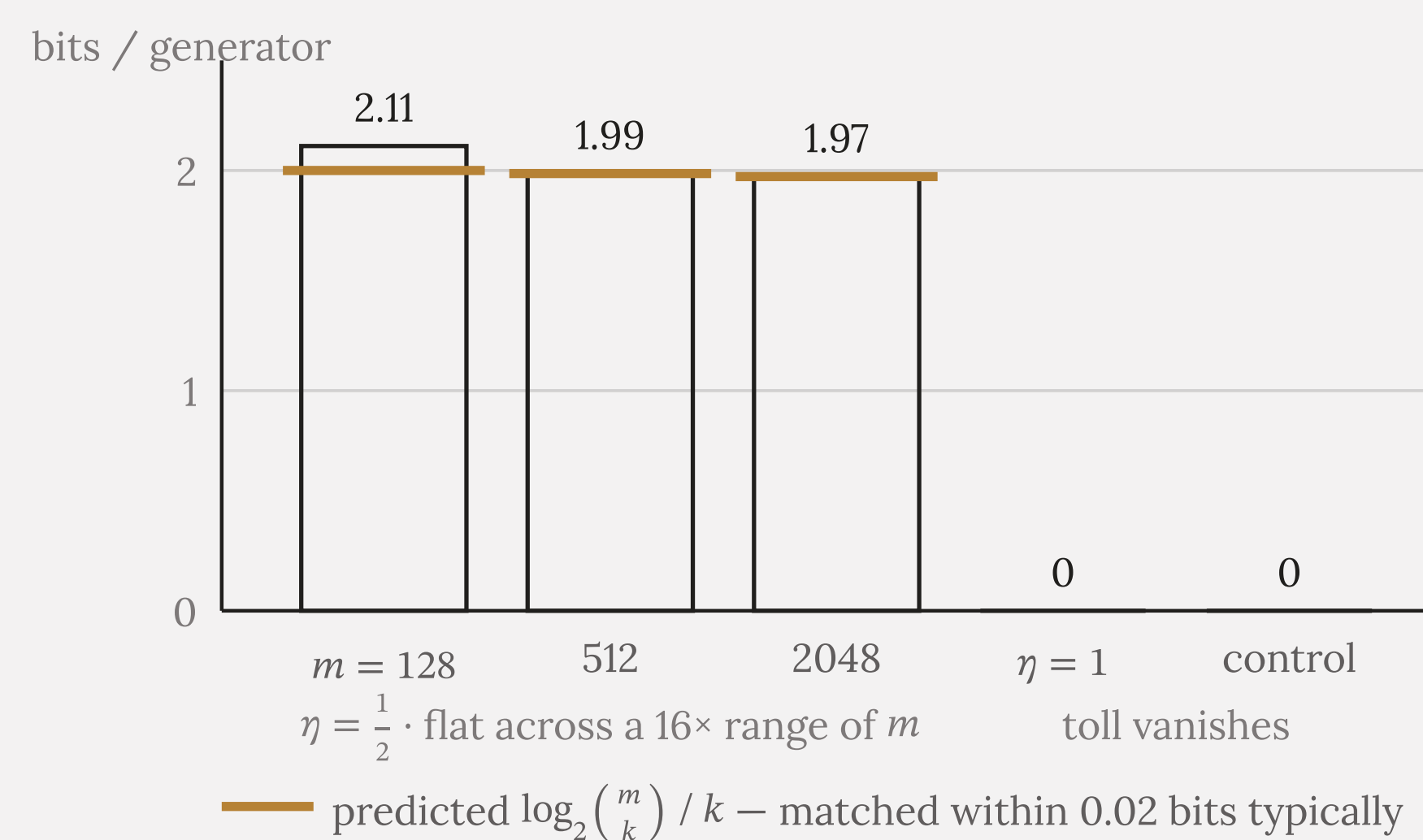
A_p is committed output, S the serialized crossing state, and $K_\epsilon = \Theta(1/\epsilon)$ the number of distinguishable settings on the Ross-Selinger synthesis grid. The two cannot both be small. The bound is carried by the schedule alone, so the folded compile-time presentation of randomized compiling inherits it verbatim — nothing depends on tracking Pauli frames.



Everything below the solid line is unreachable for a compiler fed the dispersed stream; a block compiler reaches the line up to a constant factor.

04 Not at par

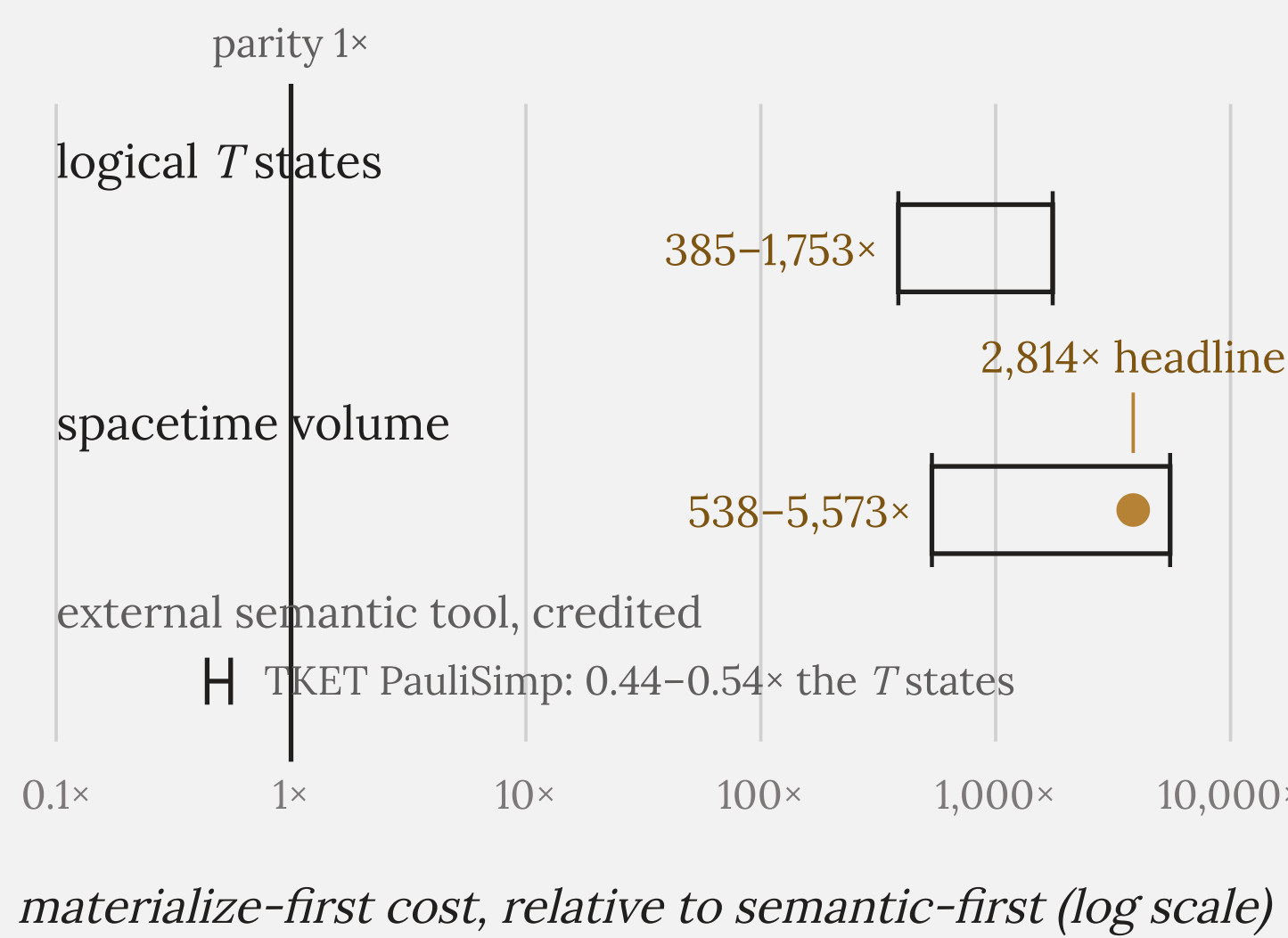
Once a committed record is *executed* rather than buffered, the committed channel pays a further $\log_2 \binom{m}{k}$ — the entropy of the compiler's own decision about which coordinates to commit early. A compiler that defers is handed that decision for free by the stream that follows.



Measured on an instrumented device under a preregistration frozen before the run. The toll is exactly zero when the commitment schedule is public.

05 What it costs downstream

Propagating one fixed total error budget through actual Clifford+ T synthesis and surface-code scheduling turns the representational gap into a fault-tolerant one. Over all 96 matched target / error / allocation / QEC / factory settings, no reversal occurs.



Outputs of checked angle synthesis and the disclosed scheduling models — not universal hardware predictions.

THE EXCHANGE RATE, IN MINIATURE

One QAOA cost layer, 3-regular MaxCut on 22 vertices — $m = 33$ edges at $\epsilon = 10^{-3}$. The layer is worth 252 bits, about **31 bytes**. A compiler that defers carries those 31 bytes across every cut. One that commits early buys the same information back at roughly 30 T gates per corrected coordinate: order 10^3 **magic states**.

06 Where the bound bites

Two hypotheses hold

Variational and Trotter cost layers: rounds contiguous in the stream, coefficients free to vary round to round.

One fails

Controlled-power ladders and phase estimation: contiguous, but round one already fixes the aggregate.

Clifford records — routing, swap, and the Pauli-frame insertions of randomized compiling itself — sit interior to the block and normalize away, so the sign-randomized and folded presentations obey the same floor.

1,080 matched cells, 9 representations × 5 compilers	1,728 external-baseline cells, bridge loss reported apart
288 fixed-total-error fault-tolerant estimates	21,060 workload and negative-control cells, six-factor ablation

Every headline number is generated from a normalized frozen registry of 24,748 records; one command reruns the data audits, tables, figures and tests, and an independent clean-room directory reconstructs the theorem, benchmark and resource-estimate subsets.

Full paper, artifacts and preregistrations: [link or contact here](#)

T A K E A W A Y

Randomize and lower late.

The information a compiler discards before aggregating cannot be recovered for free — it must be bought back in memory or in magic states, and buying it back early is strictly the more expensive of the two. Treat recoverability as a controlled variable in compilation and resource estimation.

FUNDING

This work was supported in part by the National Natural Science Foundation of China under Grant No. 62476209, in part by the Key Research and Development Program of Shaanxi under Grant No. 2024CY2-GJHX-18, and in part by the Fundamental Research Funds for the Central Universities under Grant No. QTZX26097, the Science, Technology and Innovation Commission of Shenzhen Municipality (Grant Nos. JCYJ20170412152620376 and KYTDP20181011104202253), the Innovation Program for Quantum Science and Technology (2021ZD0301703), Guangdong Major Project of Basic Research (2025B0303000007), and the Shenzhen Science and Technology Program (Grant No. KQTD20200820113010023).